

Deterministic Optimization

Algebraic Aspects of Linear Optimization

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Basic Solution in Standard Form

LP

Algebraic Aspects of Linear Optimization

Learning Objectives for this lesson

- Apply the concept of basic solution to standard form LP

Standard Form LP

- A **Standard Form** linear program is written as
 - $\min c^\top x$
 - s.t. $Ax = b$
 $x \geq 0$
 - $x \in R^n$, i.e. there are n variables
 - $A \in R^{m \times n}$, i.e. there are m equality constraints
 - We always assume all the m equality constraints are linearly independent
 - Because otherwise we can remove all linearly dependent constraints.

Basic Solution in Standard Form

- A basic solution is the unique solution to n linearly independent active constraints.
- For a standard form LP, we already have m linearly independent active constraints.
- Need $n - m$ additional linearly independent active constraints
- Where to find them?
- Only from nonnegative constraints: $x_i \geq 0$
- But which to choose to make active?

Procedure to Find A Basic Solution in Standard Form LP

- Since $Ax = b$ has m linearly independent rows, there are also m linearly independent columns
- Choose m such linearly independent columns, denote the corresponding $m \times m$ matrix as B , called **basis matrix**. The corresponding m x_i 's are denoted as x_B , called **basic variables**
- So A is partitioned as $A = [B, N]$, where N is $m \times (n - m)$. The corresponding $(n - m)$ x_i 's are denoted as x_N , **nonbasic variables**
- Choose $x_i = 0$ for all i corresponds to the columns in N , $x_N = 0$

Why does this method work?

- We can write the n active constraints as

- $$\begin{bmatrix} B & N \\ 0 & I \end{bmatrix} \begin{bmatrix} x_B \\ x_N \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix}$$

- Since B is an invertible matrix, and I is identity matrix, the whole matrix is invertible, therefore, n active constraints are linearly independent
- Thus, there is only one solution, which is a basic solution
- The solution can be computed:
 - $Bx_B = b \Rightarrow x_B = B^{-1}b$
 - $x_N = 0$

Summary

- We learned:
 - How to find n linearly independent active constraints in standard form problem